

Worksheet 3D: Graphing Quadratics

Annotated Revision Guide

AIML Foundations Mathematics Dublin and Dún Laoghaire ETB

Instructor: Josh Aaron

Parts A–B: The Basic Parabola and Vertical Stretch/Compression (Problems 1–16)

Concept. The simplest parabola is $f(x) = x^2$: vertex at origin, symmetric about $x = 0$, opens upward. The coefficient a in $y = ax^2$ controls two things: **direction** (sign of a) and **width** ($|a|$ large $\Rightarrow$ narrow; $|a|$ small $\Rightarrow$ wide).

Method. Build a table of $(x, f(x))$ values for $x = -3, -2, -1, 0, 1, 2, 3$. Plot and connect smoothly. To compare widths: for the same x , $y = 2x^2$ gives twice the y -value of $y = x^2$.

Watch out. $a > 0$ opens up (minimum). $a < 0$ opens down (maximum). $|a|$ large does *not* mean the parabola is tall — it means it is *narrow* (reaches the same height at a smaller $|x|$).

Where this leads. The factor a becomes the coefficient a in vertex form $y = a(x - h)^2 + k$. It determines whether the vertex is a min ($a > 0$) or max ($a < 0$).

Explore further. GeoGebra ([geogebra.org/calculator](https://www.geogebra.org/calculator)): type $y = ax^2$ and create a slider for a to see how changing a deforms the parabola in real time.

1. Complete the table for $f(x) = x^2$.

Values. $x = -3, -2, -1, 0, 1, 2, 3$: $f(x) = 9, 4, 1, 0, 1, 4, 9$.

3. Properties of $f(x) = x^2$.

- (a) **Vertex.** $(0, 0)$.
- (b) **Symmetric about.** $x = 0$ (the y -axis).
- (c) **Opens.** Upward.
- (d) **y -intercept.** $f(0) = 0$.

4. $f(x) = x^2$ and the x -axis.

- (a) $x^2 = 0$. One solution: $x = 0$.
- (b) $x^2 = -1$. No real solutions: squaring a real number cannot give a negative result.

5. Complete the table for $g(x) = -x^2$.

Values. $g(x) = -9, -4, -1, 0, -1, -4, -9$.

6. How does $g(x) = -x^2$ compare to $f(x) = x^2$?

Answer. Reflected over the x -axis. Every y -value is negated.

7. Properties of $g(x) = -x^2$.

Vertex. $(0, 0)$.

Opens. Downward.

Maximum. $g(0) = 0$.

8. Complete the rule for $y = ax^2$.

Rule. $a > 0$: opens **up**. $a < 0$: opens **down**.

9. Compare $f(x) = x^2$ and $g(x) = 2x^2$.

f : 4, 1, 0, 1, 4. g : 8, 2, 0, 2, 8.

Effect. g doubles every y -value: narrower parabola.

13. Which is narrower: $y = x^2$ or $y = 3x^2$?

Answer. $y = 3x^2$: at $x = 1$, $y = 3$ vs $y = 1$. Narrower.

14. Which is wider: $y = x^2$ or $y = 0.25x^2$?

Answer. $y = 0.25x^2$: at $x = 2$, $y = 1$ vs $y = 4$. Wider.

15. Match equations to descriptions.

$y = 5x^2$. B: very narrow, opens up.

$y = 0.1x^2$. C: very wide, opens up.

$y = -x^2$. A: same shape as $y = x^2$, flipped.

$y = -3x^2$. D: narrow, opens down.

Part C: Vertical Shifts (Problems 17–24)

Concept. Adding a constant k to $y = x^2$ shifts the parabola: $y = x^2 + k$ moves the vertex to $(0, k)$. The shape is identical; only the position changes.

Method. Vertex of $y = x^2 + k$ is $(0, k)$. Number of x -intercepts: if $k > 0$, parabola is entirely above x -axis (0 intercepts); $k = 0$, vertex on x -axis (1 intercept); $k < 0$,

parabola crosses x -axis (2 intercepts).

Watch out. $y = x^2 + k$ shifts *vertically*. Horizontal shifts are handled in Part D with $(x - h)^2$.

Where this leads. Vertical and horizontal shifts combine in vertex form $y = a(x - h)^2 + k$: the vertex is at (h, k) , not $(0, 0)$.

Explore further. GeoGebra: type $y = x^2 + k$ and add a slider for k .

17. Compare $y = x^2$, $y = x^2 + 3$, $y = x^2 - 2$ at $x = -2, -1, 0, 1, 2$.

f : 4, 1, 0, 1, 4.

$g(+3)$: 7, 4, 3, 4, 7.

$h(-2)$: 2, -1, -2, -1, 2.

18. $y = x^2 + 3$ vs $y = x^2$.

Answer. Shifted **up** 3 units.

19. $y = x^2 - 2$ vs $y = x^2$.

Answer. Shifted **down** 2 units.

20. Vertices of $y = x^2$, $y = x^2 + 5$, $y = x^2 - 4$, $y = x^2 + 100$.

Vertices. $(0, 0)$; $(0, 5)$; $(0, -4)$; $(0, 100)$.

22. Equations for vertex at $(0, 7)$ and $(0, -3)$.

(a) $y = x^2 + 7$.

(b) $y = x^2 - 3$.

23. How many x -intercepts?

(a) $y = x^2 + 1$ 0: entirely above x -axis.

(b) $y = x^2$ 1: vertex touches x -axis.

(c) $y = x^2 - 4$ 2: crosses at $x = \pm 2$.

24. Solve $x^2 - 4 = 0$. Relate to graph of $y = x^2 - 4$.

Solve. $x = \pm 2$.

Graph. These are the x -intercepts of $y = x^2 - 4$.

Part D: Horizontal Shifts (Problems 25–32)

Concept. $y = (x - h)^2$ shifts the parabola *horizontally*. The vertex moves to $(h, 0)$. The shift is in the **opposite** direction to the sign: $(x - 2)^2$ shifts *right* by 2; $(x + 3)^2$ shifts *left* by 3.

Method. To find the vertex, identify h from $(x - h)^2$. Remember: $y = (x + 3)^2 = (x - (-3))^2$ means $h = -3$, so vertex is $(-3, 0)$.

Watch out. The sign confusion is the number one error: "+3 inside means vertex at -3." Think of it as: what value of x makes the bracket zero? $(x + 3) = 0 \Rightarrow x = -3$.

Where this leads. Combining with vertical shifts: $y = (x - 2)^2 + 5$ has vertex at $(2, 5)$. Part E assembles these pieces into vertex form.

Explore further. Experiment in GeoGebra: $y = (x - h)^2$ with slider h .

25. Table for $f(x) = (x - 2)^2$ at $x = 0, 1, 2, 3, 4$.

$(x - 2)$: $-2, -1, 0, 1, 2$.

$f(x)$: $4, 1, 0, 1, 4$.

26. Vertex of $y = (x - 2)^2$.

Answer. $(2, 0)$.

28. Vertex of $y = (x + 3)^2$.

Answer. $(-3, 0)$: the bracket $(x + 3) = 0$ when $x = -3$.

29. Shift direction.

$(x - 2)^2$: **Right** 2 units.

$(x + 3)^2$: **Left** 3 units.

30. Vertices: (a) $(x - 5)^2$ (b) $(x + 1)^2$ (c) $(x - 10)^2$ (d) $(x + 7)^2$.

Vertices. $(5, 0)$; $(-1, 0)$; $(10, 0)$; $(-7, 0)$.

31. Equations for vertices $(4, 0)$ and $(-2, 0)$.

(a) $y = (x - 4)^2$.

(b) $y = (x + 2)^2$.

Part E: Vertex Form (Problems 33–42)

Concept. Vertex form: $y = a(x - h)^2 + k$. Vertex is (h, k) . Direction: $a > 0$ up, $a < 0$ down. Width: $|a| > 1$ narrow, $|a| < 1$ wide. This form immediately reveals all key features of a parabola.

Method. Reading off: h is from $(x - h)$; k is the constant added. Writing: start with $y = a(x - h)^2 + k$ and fill in a, h, k . To find a from a point and vertex: substitute coordinates, solve.

Watch out. $y = -0.5(x + 2)^2 - 1$: $h = -2$ (not $+2$!), $k = -1$, $a = -0.5 < 0$ so opens down.

Where this leads. Vertex form is derived from standard form by **completing the square**. This technique is the algebraic heart of Part F.

Explore further. GeoGebra (geogebra.org/m/BUD9HAPR) — vertex form explorer with sliders.

33. $y = 2(x - 1)^2 + 3$: read off all features.

$a = 2$, $h = 1$, $k = 3$. Vertex $(1, 3)$. Opens up. Narrower than $y = x^2$.

34. $y = -0.5(x + 2)^2 - 1$: read off all features.

$a = -0.5$, $h = -2$, $k = -1$. Vertex $(-2, -1)$. Opens down. Wider than $y = x^2$.

35. Write vertex form for given conditions.

(a) **Vertex $(3, 2)$, up, standard width.** $y = (x - 3)^2 + 2$.

(b) **Vertex $(-1, 4)$, down.** $y = -(x + 1)^2 + 4$.

(c) **Vertex $(0, -5)$, up, twice as narrow.** $y = 2x^2 - 5$.

(d) **Vertex $(2, -3)$, down, half as steep.** $y = -\frac{1}{2}(x - 2)^2 - 3$.

37. Sketch $y = (x - 2)^2 - 4$.

Vertex. $(2, -4)$.

Opens. Up.

y-int. $x = 0$: $(0 - 2)^2 - 4 = 4 - 4 = 0$.

x-ints. $(x - 2)^2 = 4 \Rightarrow x - 2 = \pm 2 \Rightarrow x = 0$ or $x = 4$.

38. Sketch $y = -(x + 1)^2 + 9$.

Vertex. $(-1, 9)$.

Opens. Down.

y-int. $x = 0$: $-(1)^2 + 9 = 8$.

x-ints. $(x + 1)^2 = 9 \Rightarrow x + 1 = \pm 3 \Rightarrow x = 2$ or $x = -4$.

39. Parabola through $(0, 1)$ with vertex $(2, 5)$.

Start. $y = a(x - 2)^2 + 5$. Substitute $(0, 1)$: $1 = 4a + 5 \Rightarrow a = -1$.

Equation. $y = -(x - 2)^2 + 5$.

40. Vertex $(-1, -4)$, passes through $(1, 0)$.

Start. $y = a(x + 1)^2 - 4$. Substitute $(1, 0)$: $0 = 4a - 4 \Rightarrow a = 1$.

Equation. $y = (x + 1)^2 - 4$.

42. Ball: $h(t) = -5(t - 3)^2 + 50$.

(a) **Max height.** $k = 50$ m.

(b) **Time.** $t = h = 3$ s.

(c) $t = 0$. $h(0) = -5(9) + 50 = -45 + 50 = 5$ m.

Part F: Converting Between Forms (Problems 43–50)

Concept. Standard form $y = ax^2 + bx + c$ and vertex form $y = a(x - h)^2 + k$ describe the same parabola. **Standard to vertex:** use $h = -b/(2a)$, then compute $k = f(h)$; or **complete the square**. **Vertex to standard:** expand and simplify.

Method. Completing the square for $y = x^2 + bx + c$: add and subtract $(b/2)^2$:
 $y = (x + b/2)^2 + c - (b/2)^2$.

Watch out. When $a \neq 1$, factor out a first: $y = 2x^2 + 8x + 6 = 2(x^2 + 4x) + 6 = 2(x^2 + 4x + 4 - 4) + 6 = 2(x + 2)^2 - 2$.

Where this leads. Completing the square is also used to solve $ax^2 + bx + c = 0$ — this is how the quadratic formula is derived.

Explore further. The two forms are analogous to $y = mx + b$ (slope-intercept) vs $y - y_1 = m(x - x_1)$ (point-slope) for lines: same line, different emphasis.

43. $y = x^2 - 6x + 5$: find vertex.

$a = 1, b = -6, c = 5$. $h = -(-6)/(2) = 3$.

$k = f(3)$. $9 - 18 + 5 = -4$.

Vertex. $(3, -4)$.

44. $y = x^2 + 4x + 3$.

Vertex. $h = -2$; $k = 4 - 8 + 3 = -1$. Vertex $(-2, -1)$.

y-int. $c = 3$.

x-ints. $(x + 1)(x + 3) = 0 \Rightarrow x = -1, -3$.

45. $y = -x^2 + 2x + 8$.

Vertex. $h = 1$; $k = -1 + 2 + 8 = 9$. Vertex $(1, 9)$.

Max or min. $a = -1 < 0$: **maximum** of 9.

x-ints. $-x^2 + 2x + 8 = 0 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x - 4)(x + 2) = 0 \Rightarrow x = 4, -2$.

46. $y = 2x^2 + 8x + 6$.

Vertex. $h = -2$; $k = 8 - 16 + 6 = -2$. Vertex $(-2, -2)$.

Factor. $2(x^2 + 4x + 3) = 2(x + 1)(x + 3) = 0 \Rightarrow x = -1, -3$.

47. Convert to vertex form: (a) $y = x^2 + 6x + 10$ (b) $y = x^2 - 2x - 3$.

(a) $(x^2 + 6x + 9) + 10 - 9 = (x + 3)^2 + 1$.

(b) $(x^2 - 2x + 1) - 3 - 1 = (x - 1)^2 - 4$.

48. Convert: (a) $y = x^2 + 8x + 20$ (b) $y = x^2 - 10x + 21$.

(a) $(x + 4)^2 + 4$.

(b) $(x - 5)^2 - 4$.

49. Vertex to standard: (a) $y = (x + 2)^2 - 5$ (b) $y = 3(x - 1)^2 + 4$.

(a) $x^2 + 4x + 4 - 5 = x^2 + 4x - 1$.

(b) $3(x^2 - 2x + 1) + 4 = 3x^2 - 6x + 7$.

50. Vertex to standard: (a) $y = -(x - 4)^2 + 10$ (b) $y = 2(x + 3)^2 - 7$.

(a) $-(x^2 - 8x + 16) + 10 = -x^2 + 8x - 6$.

(b) $2(x^2 + 6x + 9) - 7 = 2x^2 + 12x + 11$.

Part G: Discriminant and Roots — Graphical View (Problems 51–60)

Concept. The discriminant $\Delta = b^2 - 4ac$ predicts the number of x -intercepts: $\Delta > 0$: two roots (parabola crosses x -axis twice); $\Delta = 0$: one repeated root (vertex

on x -axis); $\Delta < 0$: no real roots (parabola entirely above or below x -axis). This links the *algebraic* discriminant to the *geometric* position of the parabola.

Method. Compute $b^2 - 4ac$ first. Then if $\Delta \geq 0$, solve with the quadratic formula. If $\Delta < 0$, stop: no real solutions.

Watch out. A parabola opening downward with $\Delta < 0$ lies entirely *below* the x -axis. A parabola opening upward with $\Delta < 0$ lies entirely *above*. The direction (a) matters when interpreting the geometry.

Where this leads. Complex roots when $\Delta < 0$: $x = \frac{-b \pm i\sqrt{|\Delta|}}{2a}$. Worksheet 4 revisits this in the context of characteristic equations of differential equations.

Explore further. The discriminant generalises to higher-degree polynomials and is deeply connected to the *resultant* in algebraic geometry.

51. Discriminant and x -intercepts for each.

(a) $y = x^2 - 4x + 3$. $\Delta = 16 - 12 = 4 > 0$. Two intercepts ($x = 1, 3$).

(b) $y = x^2 - 4x + 4$. $\Delta = 16 - 16 = 0$. One intercept ($x = 2$).

(c) $y = x^2 - 4x + 5$. $\Delta = 16 - 20 = -4 < 0$. No intercepts.

53. $y = x^2 + bx + 9$: values of b for each case.

(a) **Two intercepts.** $\Delta = b^2 - 36 > 0 \Rightarrow |b| > 6$.

(b) **One intercept.** $\Delta = 0 \Rightarrow b = \pm 6$.

(c) **No intercepts.** $\Delta < 0 \Rightarrow |b| < 6$.

54. $y = x^2 - 6x + k$ touches x -axis exactly once: find k .

$\Delta = 0$. $36 - 4k = 0 \Rightarrow k = 9$.

55. $y = ax^2 + 4x + 1$: how many x -intercepts?

$\Delta = 16 - 4a$.

$a = 1$: $\Delta = 12 > 0$. Two.

$a = 4$: $\Delta = 0$. One.

$a = 5$: $\Delta = -4 < 0$. None.

56. $x^2 + 2x + 5 = 0$ has no real solutions. Show graphically.

Vertex. $h = -1$; $k = 1 - 2 + 5 = 4$. Vertex $(-1, 4)$.

Conclusion. Vertex is at $y = 4 > 0$, parabola opens up: entirely above x -axis.

58. Vertex $(3, -4)$, $a = 1$.

- (a) **Equation.** $y = (x - 3)^2 - 4$.
 (b) **Crosses x -axis?** $\Delta = a > 0$ and vertex below axis: yes.
 (c) **Intercepts.** $(x - 3)^2 = 4 \Rightarrow x = 1$ or $x = 5$.

59. Vertex $(2, 5)$, $a = 1$.

- (a) **Equation.** $y = (x - 2)^2 + 5$.
 (b) **Crosses x -axis?** Vertex at $y = 5 > 0$, opens up: no x -intercepts.

60. Vertex $(-1, 3)$, $a = -2$.

- (a) **Equation.** $y = -2(x + 1)^2 + 3$.
 (b) **Crosses x -axis?** Opens down from $y = 3 > 0$: yes, two intercepts.
 (c) **Intercepts.** $-2(x + 1)^2 + 3 = 0 \Rightarrow (x + 1)^2 = 3/2 \Rightarrow x = -1 \pm \sqrt{3/2} = -1 \pm \frac{\sqrt{6}}{2}$.

Part H: Real-World Graphing Applications (Problems 61–66)

Concept. These problems apply vertex form to find maximum or minimum values in realistic contexts: maximum height, maximum profit, maximum area. The vertex gives the answer directly: x_{vertex} is when the optimum occurs, y_{vertex} is the optimum value.

Method. Write the quadratic. Find the vertex with $h = -b/(2a)$ or by completing the square. Verify units and physical interpretation.

Watch out. Always check that the optimum is physically feasible: negative time, negative width, or price beyond market range should be discarded.

Where this leads. Every optimisation problem in calculus has the same structure: $f'(x) = 0$ gives the critical points. For quadratics, $f'(x) = 2ax + b = 0$ gives $x = -b/(2a)$ — identical to the vertex formula.

Explore further. The fundamental theorem of calculus (Worksheet 4) shows that finding the vertex is equivalent to setting the derivative to zero.

61. $h(t) = -5t^2 + 20t + 1$.

- (a) **Vertex form.** $t^* = 2$ s; $k = h(2) = -20 + 40 + 1 = 21$. $h = -5(t - 2)^2 + 21$.
 (b) **Max height.** 21 m.
 (c) **Time.** $t = 2$ s.
 (e) **Lands.** $-5t^2 + 20t + 1 = 0$; disc = $400 + 20 = 420$; $t = (20 + \sqrt{420})/10 \approx 4.05$ s.

62. Profit: $P(x) = -2x^2 + 24x - 54$ (\$000), x = hundreds of units.

- (a) **Vertex.** $x^* = -24/(-4) = 6$. $P(6) = -72 + 144 - 54 = 18$. Max profit at 600 units.

- (b) **Sales.** 600 units.
(c) **Max profit.** \$18,000.
(d) **Break even.** $-2x^2 + 24x - 54 = 0 \Rightarrow x^2 - 12x + 27 = 0 \Rightarrow (x - 3)(x - 9) = 0 \Rightarrow x = 3$ or 9. Hundreds of units: 300 and 900.

63. Fencing: $A(w) = -2w^2 + 200w$.

- (a) **Max width.** $w^* = 200/4 = 50$ m.
(b) **Max area.** $A(50) = -5000 + 10000 = 5000$ m².

64. Revenue: $R(p) = -25p^2 + 1000p$.

- (a) **Max price.** $p^* = 1000/50 = 20$ €.
(b) **Max revenue.** $R(20) = -25(400) + 20000 = -10000 + 20000 = \pounds 10000$.
(c) **Revenue zero.** $p(1000 - 25p) = 0 \Rightarrow p = 0$ (free) or $p = 40$ (no buyers). The product has zero demand above €40.

65. Suspension bridge cable: $y = 0.01x^2$.

- (a) **At $x = 50$.** $y = 0.01 \times 2500 = 25$ m.
(b) **Towers at $x = \pm 50$.** $y = 25$ m above the lowest point.

66. Fountain: $y = -0.5x^2 + 2x$.

- (a) **Max height.** $x^* = 2$ m; $y(2) = -2 + 4 = 2$ m.
(b) **Range.** $-0.5x^2 + 2x = 0 \Rightarrow x(2 - 0.5x) = 0 \Rightarrow x = 0$ or $x = 4$ m.